

DFSMSHsm Fast Replication Part 2

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Agenda



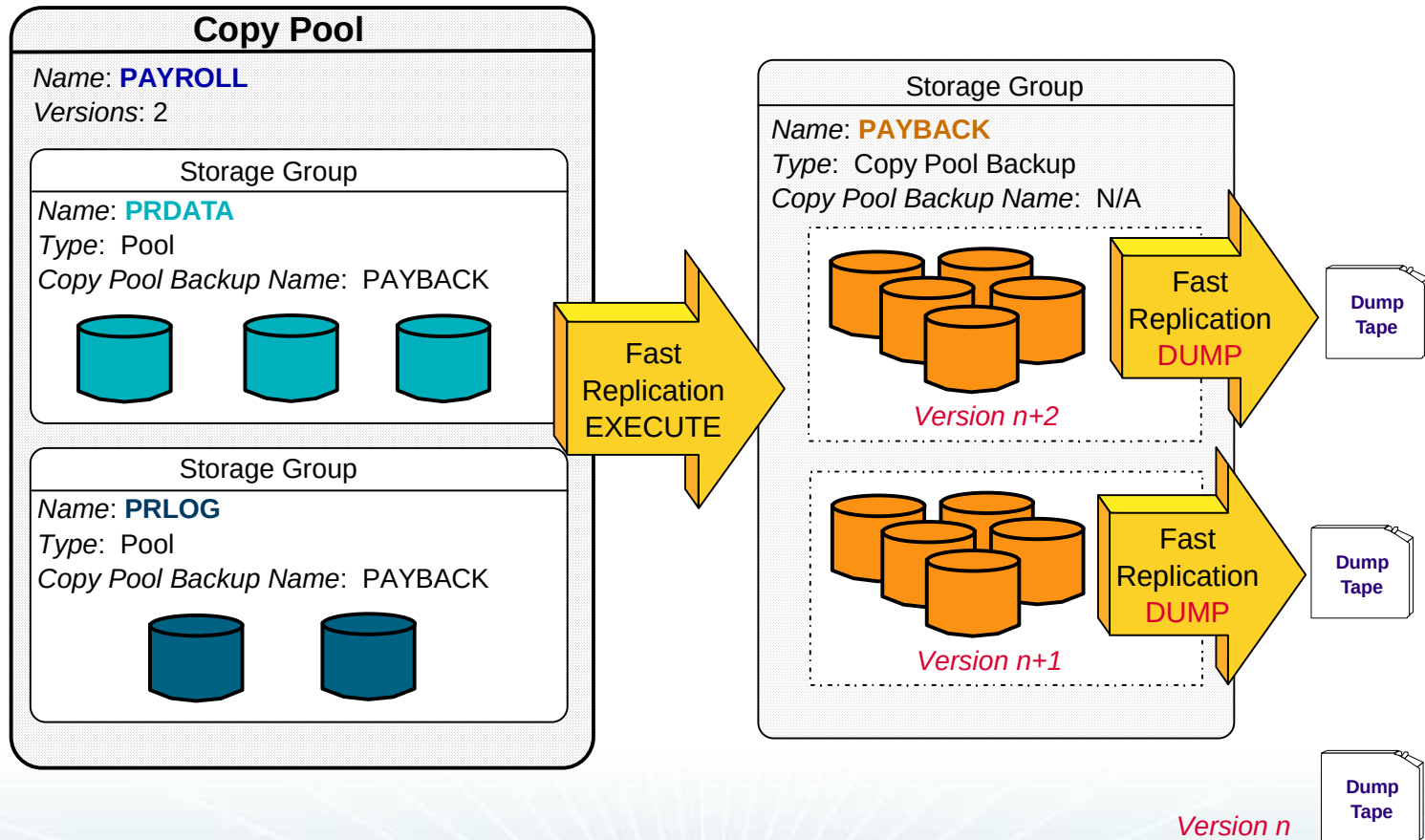
- Fast Replication Tape Support Overview
- Setup
- Creating a Tape Copy
- Recovering a Backup Copy
- Fast Replication Support Functions
- Summary

Fast Replication Tape Support Overview



- DFSMShsm manages copying FlashCopy target volumes to tape
 - By command, immediately after FlashCopy version is created
 - By command, some time after the FlashCopy Version is created
 - During Automatic Dump window
- Tape copies are 'Dump' copies
 - DFSMSdss Full-volume physical dump
 - Existing DFSMShsm Dump function used to create copies
- Maintain up to 85 versions
- Up to 5 copies per version
- Support available on z/OS V1R8

Creating a Fast Replication Tape Copy



Setup for Fast Replication Tape Copy

- SMS copy pool definition
 - Allows five dump class names to be specified
 - One for each copy to be created
 - Specify if Automatic Dump should be performed
- Dump classes are existing DFSMSHsm construct that define the attributes for the dump copy
 - **Name** – 1 to 8 characters
 - **Frequency** – When this copy should be created
 - **Expiration** – How long to keep the copy
 - **Stacking** – How many source volumes are put on a single tape
 - Up to 99
 - **Encryption** – Whether or not software encryption should be used

Setup for Fast Replication Tape Copy

(continued)



- **Dump Class Example**

- Name: **ONSITE** – Copy that is kept onsite
Frequency: **0** – Create as often as necessary
Expiration: **NOLIMIT** – Let copy roll-off
Stacking: **99** – 99 volumes on single tape
Encryption: **Yes** – Use software encryption on tape

- Name: **OFFSITE** – Copy that is taken offsite
Frequency: **0**
Expiration: **NOLIMIT**
Stacking: **99**
Encryption: **Yes**

- **Dump Class values that are not applicable with fast replication**

- **RESET** – Since dump is created from target volume, DSCHA indicator cannot be reset
- **DATASETRESTORE** – Can restore individual data sets regardless of this setting
- **VTOCCOPIES** – Not needed

★ **Recommendation** – Define new dump classes for fast replication

Creating Fast Replication Tape Copy by Command



- **FRBACKUP COPYPOOL(cpname) DUMP**
 - After FlashCopy relationships are successfully established for every source volume, DFSMShsm dumps the *target* volumes to tape
 - Dump is not started if *any* of the volumes failed
 - Target volumes are DFSMSdss dump conditioned volumes
 - ★ When dump conditioned volumes are dumped to tape, the tape will look as if the source volume was dumped directly
 - ★ DFSMShsm records will show the dump tape as a dump of the *source* volume, not the target volume
 - ★ *Recovery is done directly back to the source!*
 - The *recorded* creation time for the dump tapes will be the time that the FlashCopy was created, which is the time that the data on the tape represents
 - Dump classes defined in copy pool determine copies that are created
 - Can override by specifying one to five dump class names on command
- **RETURNCONTROL(FASTREPLICATIONEND) option**
 - Control is returned to the user after the FlashCopy relationships are established, as opposed to after all dump copies complete

Creating Fast Replication Tape Copy by Command *(continued)*



- **FRBACKUP COPYPOOL(cpname) DUMPONLY**
 - Does *NOT* establish FlashCopy relationships, *ONLY* creates a dump copy of an existing disk copy
 - Use this command to
 - ★ Create the dump copy at a time other than when the FlashCopy is created
 - ★ Create additional dump copies
 - ★ Retry failed dump copies
 - Default is to dump Generation 0 – Most recent disk copy
 - **GENERATION(gennum)**
 - **VERSION(vernum)**
 - **DATE(date)**
 - **TOKEN(token)**
 - The disk copy must be Valid to issue DUMPONLY against it

Creating Fast Replication Tape Copy by Command *(continued)*



- **DUMPONLY** – Create additional copies
 - One to five dump class names may be specified with the DUMPONLY option
 - If no dump copies exist, then these dump class names override the names specified for the copy pool definition
 - If dump copies exist, then additional copies will be created if new dump class names have been specified
 - Total of five copies can be created
 - If no dump class names are specified, then dump classes defined to copy pool are used
- **DUMPONLY** – Retry failed dump copies
 - If one or more dump copies already exist for the dump version, then DUMPONLY only retries failed volumes
 - If no dump class names are specified, then all existing dump class copies will be scanned to look for failed copies
 - If one or more dump class names are specified, then only those dump class copies will be scanned

Creating Fast Replication Tape Copy by Automatic Dump



- Copy Pool definition determines if target volumes are dumped during Automatic Dump window
 - Dumps Generation(0) only
 - If Generation(0) disk copy is failed, then copy pool is not dumped
 - Copy Pool volumes are processed before storage groups and nonSMS volumes
 - If Storage Group has AutoDump(Y), then volumes will be dumped twice
 - Set AutoDump to 'N' in storage group
 - 14 hour window does not apply to copy pool dumps
 - System Affinity can be specified in copy pool definition
- Using Automatic Dump is a good way to create a single daily tape copy when multiple FlashCopies are created throughout the day

Data Integrity



- ★ Once a dump copy has started, DFSMShsm prevents the target volumes from being overlaid with a new FlashCopy until all volumes have been dumped successfully
 - Overlaying the target volumes in the middle of the dump creates a data integrity exposure. *DFSMShsm prevents this!*
 - Once the target volumes have been overlaid, it is impossible to complete a dump copy!
- **FORCE** option override
 - Invalidates any in-progress dump copies and terminates all queued dump copies
 - Enables new FlashCopy to be created when there is an incomplete dump copy
- **FRBACKUP COPYPOOL(*cpname*) WITHDRAW**
 - Invalidates any in-progress dump copies and terminates all queued dump copies

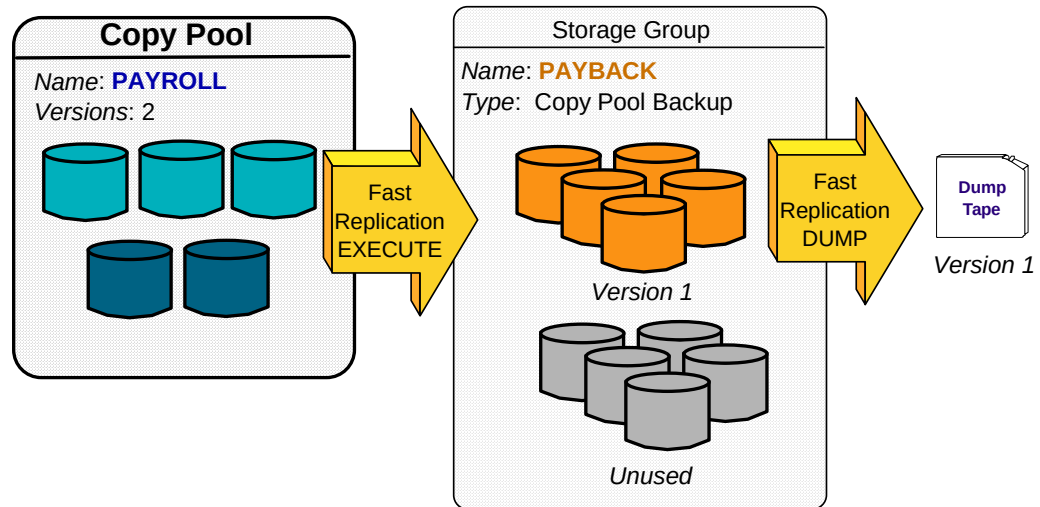
Data Integrity *(continued)*



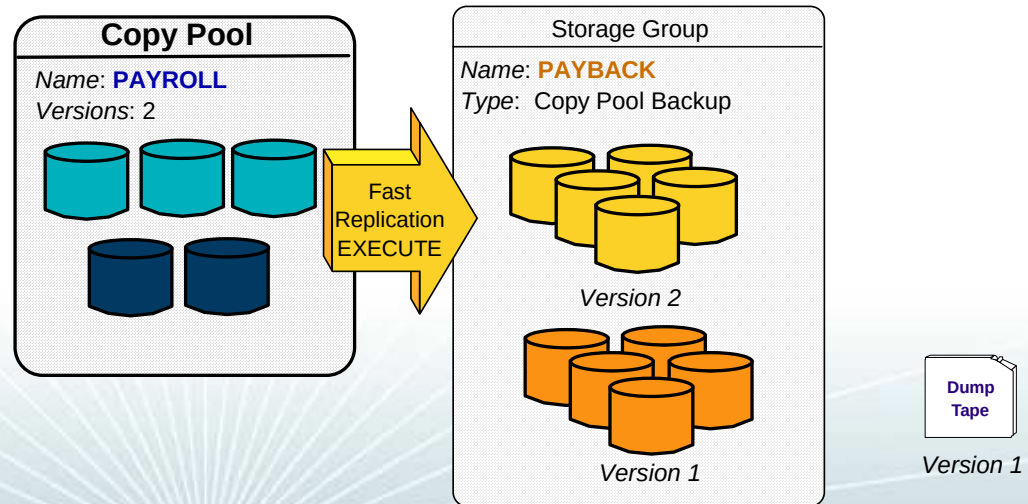
- Dump Class: **FASTREPLICATIONDUMP(REQUIRED|OPTIONAL)**
 - Enables a default override for the use of FORCE for particular dump classes
 - **REQUIRED** – Require FORCE option for incomplete dump copies
 - **OPTIONAL** – Allow new FlashCopy to be created even though not all copies have been created
 - Only applies after all dump copies have been *attempted*
- Example
 - Dump Class **ONSITE** would specify **FRDUMP(REQUIRED)** since it is the primary dump copy
 - Dump Class **OFFSITE** would specify **FRDUMP(OPTIONAL)** since it is a backup copy.
 - It could be considered more important to create the subsequent FlashCopy disk copy than to wait for the successful creation of the offsite dump copy
 - In this case, the new FlashCopy would be created if dump copy **ONSITE** had been successfully created even though dump copy **OFFSITE** had not been

Example

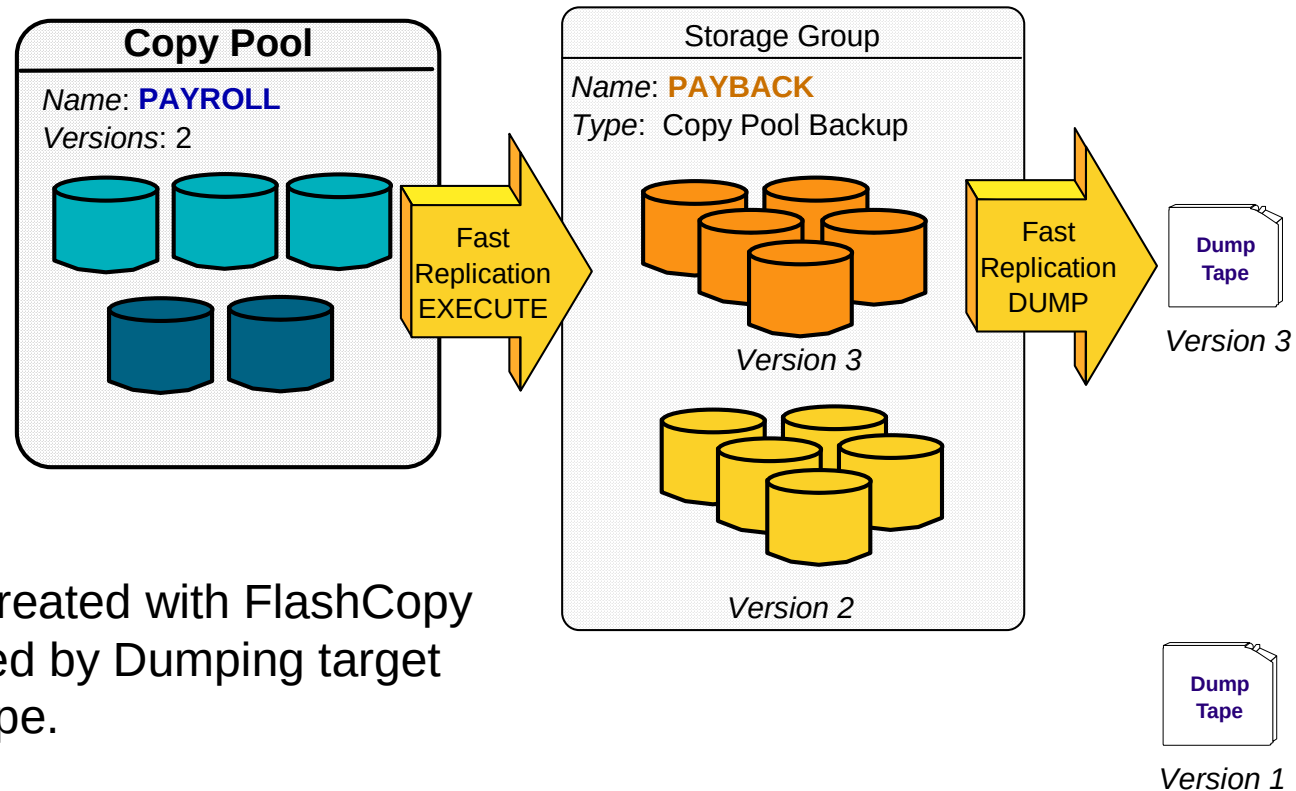
Step 1:
Version 1 is created with
FlashCopy to disk followed by
Dumping target volumes to
tape.



Step 2:
Version 2 is created with
FlashCopy to disk only.



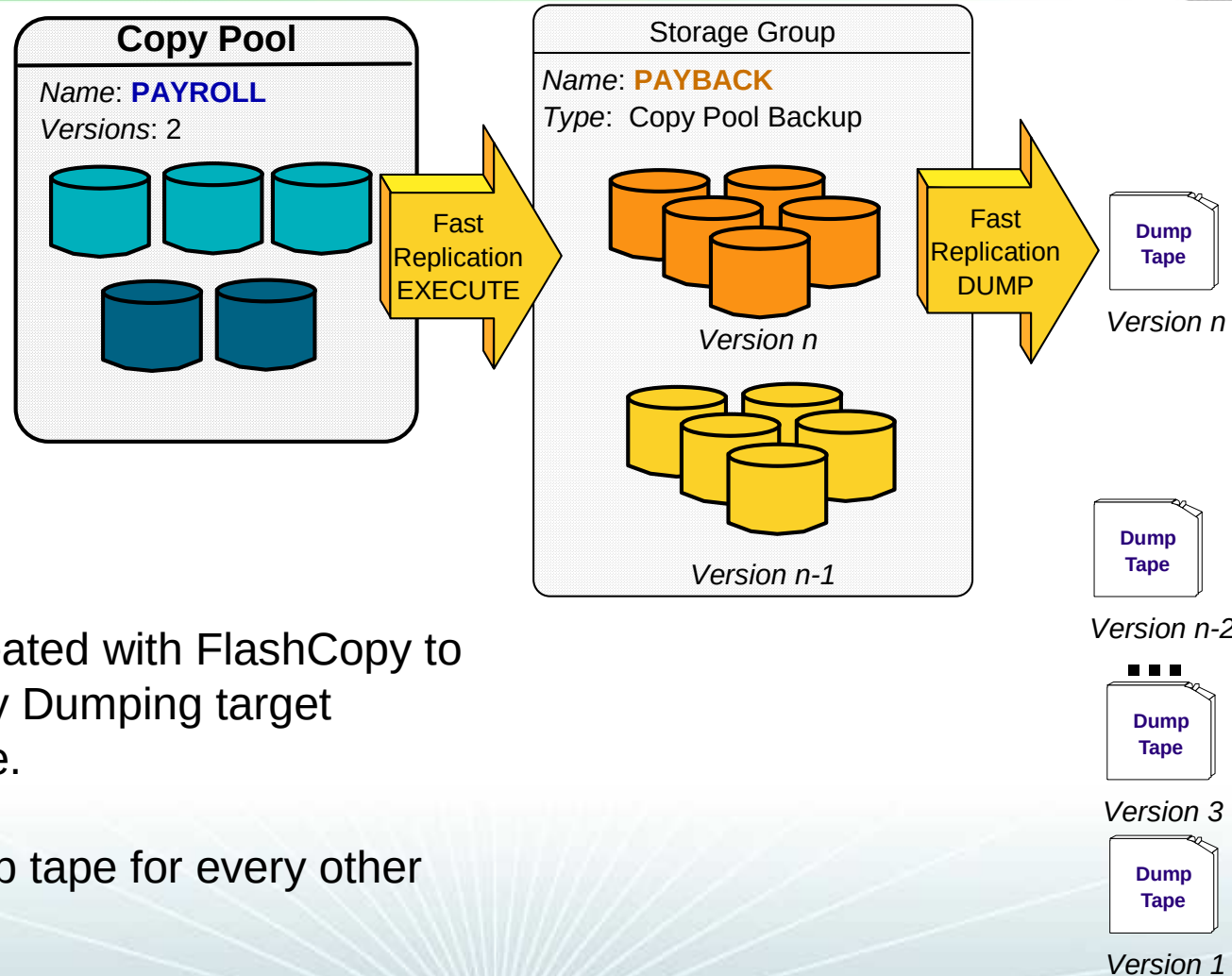
Example *(continued)*



Step 3:
Version 3 is created with FlashCopy
to disk followed by Dumping target
volumes to tape.

Notice that Version 1 disk copy was
overlaid to create Version 3

Example *(continued)*

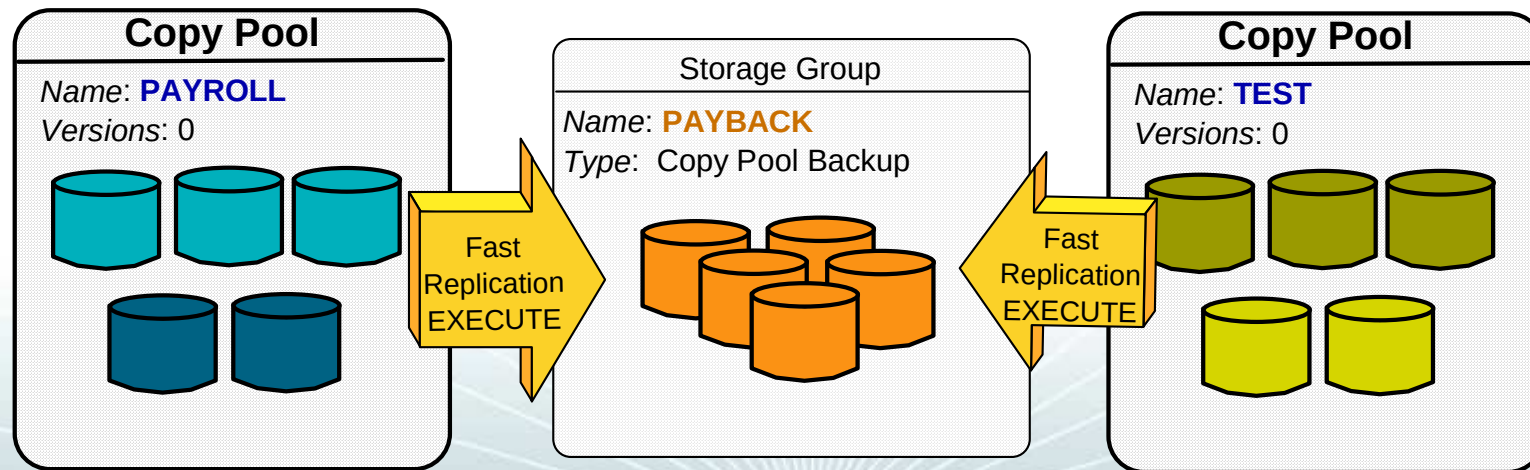


Step n :
Version n is created with FlashCopy to disk followed by Dumping target volumes to tape.

There is a dump tape for every other version

NOCOPY Mode

- Disk Versions = 0
 - NOCOPY option is used to create FlashCopy
 - DFSMSHsm Withdraws the FlashCopy relationship after the volume is successfully dumped
 - If the dump copy fails, the relationship is *not* Withdrawn, so that it can be retried
 - After the relationship is Withdrawn, the target volume can be used by a different copy pool



Fast Replication Recovery

! Restriction – Cannot recover at copy pool level from tape

- Full volume recovery from tape is still single threaded
 - This is a recognized requirement
- Create DFSMSdss jobs from LIST output (shown later) and run in parallel
- When both a disk copy and a tape copy are available, the default is to Recover from disk
(Excluding copy pools, which only recover from disk)
 - Multiple data set recoveries on the same tape are recovered with a single tape mount
- Override default with optional keywords
 - **FROMDASD** – Only recover if disk copy is available
 - **FROMDUMP** – Only recover from tape copy
 - Can specify specific dump class to use with optional **DUMPCCLASS** keyword

Fast Replication Recovery *(continued)*

- **Dump Class: FRRECOV(AVAILABLEFORMOUNT(Y|N))**
 - Enables storage administrator to specify which dump classes are mounted for recovery
 - **YES** – Copies from this dump class are available to be mounted
 - **NO** – Copies are NOT available to be mounted. Only mount when specified by command.
- **Example 1 – Normal operations**
 - Dump Class **ONSITE: FRRECOV(AFM(Y))**
 - Copies from this dump class are kept onsite and should be used when a mount is required
 - Dump Class **OFFSITE: FRRECOV(AFM(N))**
 - Copies from this dump class taken offsite and should *Not* be requested for mount
- **Example 2 – Testing is being done at Disaster Recovery site. OFFSITE tapes are needed for mounting**
 - Dump Class **ONSITE: FRRECOV(AFM(N))**
 - Copies from this dump class are Not available at the disaster recovery site. Do not mount them.
 - Dump Class **OFFSITE: FRRECOV(AFM(Y))**
 - Copies from this dump class should be requested for tape mounts.

Data Set Recovery Tasking Level

- Existing **SETSYS MAXDSRECOVERTASKS(nn)**
 - Meaning updated to specify the number of data set recoveries from *disk and tape*
 - Today, the maximum value that should be specified for this keyword is limited to the number of tape drives available
 - Default is still 2
- New **SETSYS MAXDSTAPERECOVERTASKS(nn)**
 - Specify the maximum number of data set recoveries from tape
 - Default value is the value of MAXDSRECOVERTASKS
 - Cannot exceed the value of MAXDSRECOVERTASKS
 - Keyword is new in V1R8

★ *Hint – MAXDSR should be specified before MAXDSTAPER*

Generated Requests

- The following commands generate multiple DFSMShsm work requests from a single command:
 - **FRBACKUP DUMP** – One for each volume dump
 - **FRBACKUP DUMPONLY** – One for each volume dump
 - **FRRECOV DSNAME** - One for each data set
- These generated work requests:
 - Have the same Request number as the original
 - Are displayed with QUERY output
 - Are Cancelled cumulatively via **CANCEL REQUEST(reqnum)**

RACF

- Facility Class Names – Keep Pre-V1R8 names active until all systems have migrated to V1R8

Pre V1R8	V1R8
STGADMIN.ARC.FRBACKUP	STGADMIN.ARC.FB
STGADMIN.ARC.FRBACKUP. <i>cpname</i>	STGADMIN.ARC.FB. <i>cpname</i>
STGADMIN.ARC.FRRECOV	STGADMIN.ARC.FR
STGADMIN.ARC.FRRECOV. <i>cpname</i>	STGADMIN.ARC.FR. <i>cpname</i>
STGADMIN.ARC.FRDELETE	STGADMIN.ARC.FD
STGADMIN.ARC.FRDELETE. <i>cpname</i>	STGADMIN.ARC.FD. <i>cpname</i>
None	STGADMIN.ARC.LC
None	STGADMIN.ARC.LC. <i>cpname</i>

* STGADMIN.ARC.LC protects LIST COPYPOOL command

Supporting Commands and Functions



- Holding Fast Replication functions
 - **HOLD FRBACKUP** – Hold new FRBACKUP commands
 - **HOLD FRRECOV** – Hold new FRRECOV commands
 - **TAPE** – Hold FRRECOV commands from tape
 - **DATASET** – Hold data set level FRRECOV commands
 - **HOLD DUMP(FRBACKUP)** – Hold volume dumps initiated via FRBACKUP
 - Corresponding **RELEASE** options
- Deleting unwanted Fast Replication backup versions
 - **FRDELETE COPYPOOL(cpname) VERSIONS(nn)**
 - Can delete multiple (or all) versions with single command
 - Delete only disk, only tape, or both disk and tape
 - Delete individual dump classes for a version

Supporting Commands and Functions

(continued)



- **QUERY command**
 - **QUERY COPYPOOL(*cpname*)**
 - Displays which volumes have an Active FlashCopy background copy
 - A message is issued for each disk version (new in V1R8)
 - Displays percent complete (new in V1R8)
 - **QUERY USER or REQUEST or DATASETNAME**
 - Will list each internally generated volume dump or data set recovery request generated
 - Various other Query options updated to return Fast Replication information
 - ACTIVE, SETSYS, STATISTICS, WAITING

Supporting Commands and Functions

(continued)



- **LIST command**
 - **LIST COPYPOOL**
 - Returns version information for each copy pool that has at least one attempted backup
 - **LIST COPYPOOL(*cpname*)**
 - Detailed version information for each backup version for the specified copy pool name
 - Options: FRVOLS, NOVOLS, DUMPVOLS, ALLVOLS
 - **LIST COPYPOOLBACKUPSTORAGEGROUP(*cpbsgname*)**
 - Information regarding which volumes in copy pool backup storage group that DFSMSHsm is using
 - **SELECT based on state of the version**
 - **FASTREPLICATIONSTATE**
 - *Recoverable, NonRecoverable, Failed, None*
 - **DUMPSTATE**
 - *AllComplete, RequiredComplete, Partial, None*

Supporting Commands and Functions

(continued)



- Version states
 - Disk copy (Fast Replication State)
 - **Recoverable** – Valid version that can be Flashed back
 - **NonRecoverable** – Created with NOCOPY (Versions=0)
 - **Failed** – Version did not create successfully or was withdrawn
 - **None** – No disk copy for the version
 - Tape copy (Dump State)
 - **All Complete** – Every copy of each volume was created
 - **Required Complete** – Every Required copy of each volume was created. For Optional copies, one or more volumes failed
 - **Partial** – One or more volumes failed for Required and/or Optional copies
 - **None** – No dump copies exist for the version

Recovery from Tape using DFMSdss

Use LIST COPYPOOL(cpname) to gather information

```
-- DFSMShsm CONTROL DATASET --COPY POOL--LISTING ----- AT 09:25:44 ON 06/03/07 FOR SYSTEM=3090

COPYPOOL=CP1

VERSION  VTOKENQ    DATE          TIME          FASTREPLICATIONSTATE  DUMPSTATE
 001      Y        2006/03/06    16:35:24    RECOVERABLE          ALLCOMPLETE
TOKEN(C)=C' '
TOKEN(H)=X' '
TOTAL NUM OF VOLUMES=00005

DUMPClass  REQUIRED  DUMPSTATE  VOLSSUC    EXPDATE    AVAILABLE
DCLASS1    Y        COMPLETE   00005      2006/04/05    Y

          HWCOMP  ENCRYPT  ENCTYPE  RSAKEY/KPWD
          NO      NONE      *****

SOURCE  DUMPVOLS                                DEVICE TYPE
SRC001  A00146 A00150 A00152                                3490
FILE SEQ=00, DSN=DFHSM.DMP.DCLASS1.VSRC001.D06065.T243516
SRC002  A00148                                3490
FILE SEQ=00, DSN=DFHSM.DMP.DCLASS1.VSRC002.D06065.T243516
SRC003  A00147                                3490
FILE SEQ=00, DSN=DFHSM.DMP.DCLASS1.VSRC003.D06065.T243516
SRC004  A00149                                3490
FILE SEQ=00, DSN=DFHSM.DMP.DCLASS1.VSRC004.D06065.T243516
SRC005  A00151                                3490
FILE SEQ=00, DSN=DFHSM.DMP.DCLASS1.VSRC005.D06065.T243516

----- END OF -- COPY POOL -- LISTING -----
```

Recovery from Tape using DFSMSdss

(continued)



Build DFSMSdss command from LIST output

```
//DSSREST JOB ,CLASS=A
//STEP01 EXEC PGM=ADRDSSU,REGION=4096K
//*****/
//SYSPRINT DD SYSOUT=*
//SYSUDUMP DD SYSOUT=V,OUTLIM=3000
//DBACK99 DD DISP=OLD,DSN=DFHSM.DMP.DCLASS1.VSRC002.D06065.T243516,
//          UNIT=3490,VOL=(,0,,SER=A00148)
//SYSIN DD *
        RESTORE FULL -
            INDDNAME (DBACK99) -
            PURGE COPYVOLID CANCELERROR -
            OUTDY (SRC002)
/*
```

Use these DFSMSdss jobs to run multiple volumes recoveries from tape in parallel.

Do NOT create DFSMSHsm tapes with TAPESECURITY(PASSWORD)
DFSMSdss will not have authority to mount tapes

Supporting Commands and Functions

(continued)



- **AUDIT**

- **AUDIT COPYPOOLCONTROLS**

- Scans the DFSMSHsm control records associated with copy pool processing
 - Reports errors **only**. FIX is not an option.
 - Optionally accepts a specific copy pool name

Summary



- Joint solution across DB2, DFSMS and Storage Hardware
- FlashCopy environment managed by DFSMSHsm
- Base support for automation of future FlashCopy enhancements